# Comparative Analysis of Radio Network Planning Reports

Kagal (Kinjawadekar) vs. Ralegaon (Chaudhari) vs. Pombhurna (Patil)

4G LTE Cellular Network Design — 3GPP TR 38.901 V18.0.0

Auto-Generated Comparison Report

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**Abstract**

This document presents a section-by-section comparison of three independently authored radio network planning reports for the same assignment. All three reports design a 4G LTE network for a different town in Maharashtra using 3GPP TR 38.901 V18.0.0 and then simulate small-scale fading via the Clustered Delay Line (CDL) channel model. The towns are: **Kagal** (Kolhapur district) by Aditya Kinjawadekar, **Ralegaon** (Yavatmal district) by Atharva S Chaudhari, and **Pombhurna** (Chandrapur district) by Prasad Patil (BT23ECE096). Each section highlights agreements, disagreements, and unique features across the three reports.


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# 1 Report Metadata and Formatting

Table 1: Report Metadata Comparison

| **Attribute** | **Kagal (Kinjawadekar)** | **Ralegaon (Chaudhari)** | **Pombhurna (Patil)** |
|---|---|---|---|
| Town / Area | Kagal, Kolhapur | Ralegaon, Yavatmal | Pombhurna, Chandrapur |
| Population (est.) | ≈49,000 (2026) | ≈18,500 (2026 est. from 2011 census of 13,766) | ≈20,000–25,000 (2011 census) |
| Town Area | ≈8–10 $km^2$ | 18.2 $km^2$ | ≈12 $km^2$ |
| Elevation | 553 m | 248 m | 230–250 m |
| Document Format | LaTeX 2-column, 10pt | PDF, single-column | PDF, single-column |
| Total Pages (approx.) | ∼9 (body) + appendix | 17 pages | 16 pages |
| Appendix Code? | Yes (2 Python listings) | No separate appendix (code in body) | Yes (code in body, Sec. 4) |
| Figures Included? | 6 site photos + 3 simulation plots | 2 maps | 1 site analysis map |

**Key Observations:**

- Kagal report uses professional LaTeX two-column formatting; the other two are single-column PDF reports.
- Kagal includes extensive site photography (street-level NLOS evidence); Ralegaon includes geographic/KML maps; Pombhurna includes a schematic site analysis diagram.
- All three cover the same two-part assignment: Part 1 (network planning) and Part 2 (channel matrix / CDL simulation).

# 2 Site Analysis and Location Overview

Table 2: Site Analysis Comparison

| **Feature** | **Kagal** | **Ralegaon** | **Pombhurna** |
|---|---|---|---|
| Coordinates | 16°35′N, 74°19′E | 20°25′0″N, 78°31′0″E | Near Chandrapur (approx. 20°N, 79°E) |
| Terrain Type | Semi-urban, 2–3 storey buildings | Hot tropical, cotton farming land | Flat to gently undulating, Deccan Plateau |
| Key Highway | NH-48 (6-lane, western edge) | State Highway (diagonal through town) | NH-930 (formerly SH-263, through town core) |
| Key Water Body | Dudhganga River | None prominent | Penganga River (3–5 km SW) |
| Commercial Area | Market / town centre | Main market & ginning mills | Main bazaar along NH-930 |
| Building Heights | 6–10 m (2–3 storeys) | 6–9 m (1–3 storeys) | 2–3 floors masonry (central), 1 floor (periphery) |
| **Coverage Zones** | 6 identified (Highway, River, Market, Bus Stand, Residential, Agricultural) | 4 identified (Market/Core, Ginning Mills, Highway, Agricultural) | 5 identified (Market/NH-930, Central Res., N&S Res., East Fringe, West/Penganga) |
| Visual Evidence | 4 street-level photos | KML-based geographic maps | Satellite imagery + schematic map |

**Differences:**

- **Kagal** is the most densely populated (≈49k) and smallest area (≈8–10 km$^2$), leading to a compact network.
- **Ralegaon** has the largest area (18.2 km$^2$) but smallest population (≈18.5k), resulting in a sprawling design with 13 umbrella cells.
- **Pombhurna** falls between the two in density and area.
- Kagal is the only report providing street-level photographs as propagation environment evidence.
- Ralegaon uniquely uses exact KML/GPS waypoints for site placement.

## 3 System Parameters and Link Budget

This is the most critical section for comparison, as the link budget directly determines the maximum cell radius.

Table 3: System Parameters Comparison

| **Parameter** | **Kagal** | **Ralegaon** | **Pombhurna** |
|---|---|---|---|
| Carrier Frequency $f_c$ | 1800 MHz | 1800 MHz | 1800 MHz |
| Transmit Power $P_t$ | 46 dBm | 46 dBm | 46 dBm |
| BS Antenna Gain $G_{\text{ant}}$ | 18 dBi | 18 dBi (Table 3) | 17 dBi |
| Cable / Feeder Loss | 2 dB | 2 dB | 3 dB |
| Rx Sensitivity $P_{r,\min}$ | −104 dBm (given) | −104 dBm (given) | −94 dBm (computed) |
| UE Antenna Gain | — (not used) | — (not used) | 0 dBi (explicit) |
| Body Loss | — (not used) | — (not used) | 3 dB |
| Noise Figure | — (not used) | — (not used) | 7 dB |
| Indoor Penetration Loss | — (not used) | — (not used) | 10 dB |
| Shadow Fading Margin $L_{\text{sf}}$ | 8 dB | 8 dB | 8 dB |
| BS Height $h_{\text{BS}}$ | 25 m | 25 m | 25 m |
| UT Height $h_{\text{UT}}$ | 1.5 m | 1.5 m | 1.5 m |
| Bandwidth | — (not specified) | — (not specified) | 10 MHz |
| Min SNR | — (not specified) | — (not specified) | 3 dB (QPSK 1/3) |

### 3.1 Link Budget Calculation — Critical Difference

Table 4: Link Budget Comparison — $L_{\max}$ Derivation

| **Step** | **Kagal** | **Ralegaon** | **Pombhurna** |
|---|---|---|---|
| Simple $L_{\max}$ | $46 - (-104) = 150$ dB | $46 - (-104) = 150$ dB | Not computed separately |
| Extended formula | $P_t + G_{\text{ant}} - L_{\text{cable}} - P_{r,\min} - L_{\text{sf}}$ | $P_t + G_{\text{ant}} - L_{\text{cable}} - P_{r,\min} - L_{\text{sf}}$ | $\text{EIRP} + G_{\text{UE}} - L_{\text{body}} - M_{\text{sf}} - L_{\text{pen}} - P_{r,\min}$ |
| Substitution | $46 + 18 - 2 + 104 - 8$ | $46 + 18 - 2 + 104 - 8$ | $60 + 0 - 3 - 8 - 10 - (-94)$ |
| **Result** $L_{\max}$ | **158 dB** | **158 dB** | **133 dB** |

**Critical Difference:**

- **Pombhurna uses a fundamentally different link budget approach.** It computes $P_{r,\min}$ from thermal noise floor (−104 dBm) + noise figure (7 dB) + $\text{SNR}_{\min}$ (3 dB) = −94 dBm, and includes body loss (3 dB) and indoor penetration loss (10 dB). This gives $L_{\max} = 133$ dB.
- **Kagal and Ralegaon** use $P_{r,\min} = -104$ dBm directly (given specification) and only subtract shadow fading margin, yielding $L_{\max} = 158$ dB.
- The 25 dB difference in $L_{\max}$ **drastically changes** $R_{\max}$: Pombhurna gets $R_{\max} = 843$ m vs. 3680 m for the other two. This is the single largest divergence across the three reports.

# 4 Path Loss Models

Table 5: Path Loss Model Comparison

| Aspect | Kagal | Ralegaon | Pombhurna |
|---|---|---|---|
| Primary Model | UMa NLOS | UMa NLOS | UMa NLOS |
| 3GPP Reference | TR 38.901 Table 7.4.1-1 | TR 38.901 Table 7.4.1-1 | TR 38.901 Table 7.4.1-1 |
| UMa NLOS formula | $13.54 + 39.08 \log_{10}(d_{3D}) + 20 \log_{10}(f_c) - 0.6(h_{UT} - 1.5)$ | Same | Same |
| $\sigma_{\text{SF}}$ (UMa NLOS) | 6.0 dB | Not stated | Not stated |
| UMi NLOS (microcell) | Yes, presented | Not presented | Not presented |
| RMa NLOS (umbrella) | Referenced | Used for 13 umbrella cells | Not presented |
| Breakpoint $d'_{BP}$ | 452.4 m | 452.4 m | Not computed (but implicit) |
| UMa LOS formula | Fully stated (2 regimes) | Fully stated (2 regimes) | Not stated |
| NLOS justification | Dense residential NLOS photos | Conservative worst-case | Conservative worst-case |

**Observations:**

- All three use the same UMa NLOS formula from 3GPP TR 38.901, which is correct for the assignment.
- **Kagal** is the only report that explicitly presents the UMi NLOS model for microcell planning.
- **Ralegaon** relies heavily on the RMa NLOS model for its 13 umbrella cells.
- **Pombhurna** derives a frequency-dependent delay spread formula ($\mu_{\log_{10}(\text{DS})} = -6.28 - 0.204 \log_{10}(f_c)$) from Table 7.5-6, which the other reports do not explicitly show.

# 5 Maximum Cell Radius ($R_{\max}$) and D/R Design

Table 6: $R_{\max}$ and Network Sizing Comparison

| Parameter | Kagal | Ralegaon | Pombhurna |
|---|---|---|---|
| $L_{\max}$ used | 158 dB | 158 dB | 133 dB |
| $R_{\max}$ (UMa NLOS) | 3680 m (3.68 km) | 3680 m (3.68 km) | 843 m (0.843 km) |
| $R_{\max}$ (UMi NLOS) | 4868 m | — | — |
| $R_{\max}$ (RMa NLOS) | 6258 m | 6258 m | — |
| Reuse Factor $N$ | 1 (OFDMA + ICIC) | 7 | 7 |
| $D/R$ ratio | $\sqrt{3} \approx 1.732$ | $\sqrt{21} \approx 4.583$ | $\sqrt{21} \approx 4.583$ |
| $R_{\max}$ derivation method | Analytic closed-form | Analytic closed-form | Binary search (code) |
| Verification shown? | Yes (PL at 3680 m = 158 dB) | Yes (PL at 843 m = 133 dB) | Yes (table of PL vs. distance) |

**Critical Differences:**

- Kagal uses $N = 1$ (unit frequency reuse for LTE OFDMA with ICIC), giving $D/R = \sqrt{3}$. The other two use $N = 7$, giving $D/R = \sqrt{21} \approx 4.583$. The $N = 1$ approach is more modern and appropriate for LTE, since LTE uses OFDMA and does not require traditional frequency reuse.
- Pombhurna's $R_{\max}$ is **4.4× smaller** than the other two due to the conservative link budget. This forces more cells per unit area.

- Kagal and Ralegaon agree exactly on $R_{\max}$ values for both UMa NLOS (3680 m) and RMa NLOS (6258 m).

# 6 Network Design and Iterations

Table 7: Final Network Design Comparison

| **Design Aspect** | **Kagal** | **Ralegaon** | **Pombhurna** |
|---|---|---|---|
| Number of Iterations | 2 | 2 | 2 |
| **Iteration 1** | 3 sites (gaps found: eastern residential, river-side) | 8 sites linear along highway (failed: dead zones in core) | 7 cells, varied radii (6 of 7 fail $R > R_{\max}$) |
| **Final Design:** | | | |
| Total Macrocell Sites | 5 | 4 normal + 13 umbrella = 17 | 8 normal macrocells |
| Macrocell Radius | 800 m (uniform) | 750 m (normal) | 800 m (uniform) |
| Umbrella Cells | 1 ($R \approx 2500$ m) | 13 ($R = 1000$ m) | 1 ($R = 1000$ m) |
| Microcells | 2 ($R = 200$–$300$ m) | 0 | 1 ($R = 200$ m) |
| Cell Radius Approach | Uniform $R$ for all macrocells | Two tiers: 750 m + 1000 m | Uniform $R$ for macrocells |
| PL at cell edge | 132.1 dB | 131.0 dB (normal) / 135.9 dB (umbrella) | 132.1 dB |
| Margin vs. $L_{\max}$ | 25.9 dB | 27.0 dB / 22.1 dB | 0.89 dB |
| Total coverage area | 8.31 km$^2$ | ≈39.6 km$^2$ | ≈13.3 km$^2$ |
| GPS Coordinates? | No | Yes (17 lat/lon pairs from KML) | No |

**Key Differences:**

- **Ralegaon** has the most aggressive design: 17 total sites (13 umbrella + 4 normal) to cover 18.2 km$^2$. This is architecturally unusual — most plans use 1–2 umbrella cells as an overlay, not 13.
- **Kagal** has the most compact, efficient design: 5 macrocells + 2 microcells + 1 umbrella, with a large 25.9 dB margin.
- **Pombhurna** has almost no margin (0.89 dB) due to its conservative $L_{\max} = 133$ dB and $R_{\max} = 843$ m. Any cell with $R > 843$ m would fail.
- Iteration methodology is similar across all three: start with an insufficient design, identify coverage holes/failures, add sites or reduce radii.

# 7 Microcell and Umbrella Cell Justification

Table 8: Special Cell Layer Comparison

| **Cell Type** | **Kagal** | **Ralegaon** | **Pombhurna** |
|---|---|---|---|
| **Microcells** | | | |
| Count | 2 | 0 | 1 |
| $h_{\mathrm{BS}}$ | 10 m | — | 10 m |
| Radius | 200–300 m | — | 200 m |
| Model | UMi NLOS | — | UMi NLOS |
| Location | Market area + Bus stand | — | Market / NH-930 |
| PL verification | Yes (115.3 dB at 300 m) | — | Mentioned but not computed |
| **Umbrella Cells** | | | |
| Count | 1 | 13 | 1 |
| $h_{\mathrm{BS}}$ | 35 m | 35 m | 35 m |
| Radius | ≈2500 m | 1000 m | 1000 m |
| Model | RMa NLOS | RMa NLOS | RMa NLOS |
| Purpose | NH-48 highway overlay | Highway + agricultural + suburban | Highway overlay |
| Justification depth | 1 paragraph | 2 full subsections (handover mitigation + agricultural coverage) | 1 paragraph |

**Observations:**

- Ralegaon's use of 13 umbrella cells is distinctive and extensively justified with handover failure analysis and agricultural coverage arguments.
- Kagal provides explicit path loss verification for microcells (115.3 dB at 300 m $\ll L_{\max}$).
- All umbrella cells use $h_{\mathrm{BS}} = 35$ m and RMa NLOS consistently.

# 8 Part 2: Small-Scale Fading / Channel Simulation

Table 9: Part 2 — CDL Channel Simulation Comparison

| Aspect | Kagal | Ralegaon | Pombhurna |
|---|---|---|---|
| 3GPP Reference | Section 7.5, TR 38.901 | Section 7.5, TR 38.901 | Section 7.5, TR 38.901 |
| Steps Implemented | All 12 steps | Steps 1–6 (numerical) + 7–12 (discussion) | Steps 1–6 (numerical) + 7–12 (discussion) |
| **MIMO Configuration** | | | |
| BS Antennas | 8 (4-col cross-pol) | 4 (ULA) | 4 (ULA) |
| UE Antennas | 2 | 2 | 2 |
| $H$ matrix size | $2 \times 8$ | $2 \times 4$ | $2 \times 4$ |
| Simulation Distance | 583 m (from Site 1) | 500 m | 500 m |
| Number of Clusters | 20 | 20 | 3 (for demonstration) |
| Rays per Cluster | 20 | 20 | Not specified |
| $r_\tau$ (delay scaling) | 2.3 | 2.3 (code) / 2.1 (analytic) | 2.1 |
| $C_\phi$ | 1.289 | 1.289 (code) | Not stated |
| $C_\theta$ | 1.178 | 1.178 (code) | Not stated |
| **LSP Parameters (Table 7.5-6):** | | | |
| $\mu_{\log_{10}(\mathrm{DS})}$ | $-6.955$ (in code) | $-6.955$ (in code) | $-6.332$ (frequency-dependent formula) |
| ASA $\mu$ | 1.81 | 1.81 | Not stated |
| ASD $\mu$ | 1.59 | 1.59 | Not stated |
| ZSA $\mu$ | 0.88 | 0.88 | Not stated |
| ZSD $\mu$ | 0.50 | 0.50 | Not stated |
| Polarisation | Cross-polarised (Eq. 7.5-22) | VV only (simplified) | VV only (simplified) |
| Doppler included? | Yes ($v = 3$ km/h) | Yes (discussion: $v = 50$ km/h, $f_{d,\max} = 83.3$ Hz) | Yes (discussion only) |
| **Output Results** | | | |
| SVD analysis | Not shown numerically | $\sigma_1 = 1.94$, $\sigma_2 = 0.68$, CN = 2.87 | Not shown |
| Capacity estimate | Not shown numerically | 1.82 bps/Hz at 3 dB SNR | Not shown |
| Plots | PDP, APS(AOA), $\lvert H \rvert$ heatmap | None | None |

**Key Differences:**

- Kagal implements **all 12 steps** with a full $2 \times 8$ MIMO system, cross-polarised antenna patterns (Eq. 7.5-22), Doppler, and produces PDP/APS/heatmap plots. This is the most complete implementation.
- **Ralegaon and Pombhurna** implement Steps 1–6 numerically with a $2 \times 4$ ULA, and only discuss Steps 7–12 qualitatively. Notably, Ralegaon's code and analytic sections appear to be identical to Pombhurna's code (same random seed, same variable names, same comments referencing "Pombhurna" even in the Ralegaon report).
- Pombhurna uses only $N = 3$ clusters for demonstration (vs. 20 in the other two), showing explicit hand-calculations for each step.
- The delay scaling parameter $r_\tau$ differs: Kagal uses 2.3, Pombhurna uses 2.1. The Ralegaon code uses 2.3 but the analytic section uses 2.1 (inconsistency).

# 9 Python Code Comparison

Table 10: Python Code Comparison

| **Code Aspect** | **Kagal** | **Ralegaon** | **Pombhurna** |
|---|---|---|---|
| Code files | 2 separate scripts: link budget + CDL simulator | 1 embedded script | 1 embedded script |
| Language | Python (NumPy, SciPy) | Python (NumPy) | Python (NumPy) |
| Random Seed | Not specified | 42 | 42 |
| Link Budget Code | Separate modular script | Included in main script | Included in main script |
| $R_{\max}$ method | Analytic (direct solve) | Binary search | Binary search |
| Channel Sim Code | Separate modular script | Single combined script | Single combined script |
| **Code Similarity** | Distinct, independent implementation | **Ralegaon and Pombhurna code is identical.** Same variable names, same seed(42), same comments, same structure. Ralegaon code even references "Pombhurna." | |

**Key Finding:**

- **The Python code in Ralegaon and Pombhurna reports is character-for-character identical.** The Ralegaon report's Python code still contains the string `''Pombhurna 4G Network Design''` and references "Pombhurna's semi-urban environment" – clear evidence that the code was copied from Pombhurna (or both from a common source) without adaptation.
- Kagal's code is independently written with different architecture (2 separate scripts, different variable names, modular structure, $2 \times 8$ MIMO).
- Both Ralegaon and Pombhurna compute $L_{\max} = 133$ dB and $R_{\max} = 843$ m *in their code*, yet Ralegaon's report text claims $L_{\max} = 158$ dB and $R_{\max} = 3680$ m. This is an internal inconsistency in the Ralegaon report.

## 10 Comprehensive Numerical Comparison

Table 11: All Key Numerical Values — Side-by-Side

| **Quantity** | **Kagal** | **Ralegaon** | **Pombhurna** |
|---|---|---|---|
| $f_c$ | 1800 MHz | 1800 MHz | 1800 MHz |
| $P_t$ | 46 dBm | 46 dBm | 46 dBm |
| $G_{\text{ant}}$ (BS) | 18 dBi | 18 dBi | 17 dBi |
| Cable Loss | 2 dB | 2 dB | 3 dB |
| $P_{r,\min}$ | $-104$ dBm | $-104$ dBm | $-94$ dBm |
| $L_{\text{sf}}$ | 8 dB | 8 dB | 8 dB |
| $h_{\text{BS}}$ (macro) | 25 m | 25 m | 25 m |
| $h_{\text{UT}}$ | 1.5 m | 1.5 m | 1.5 m |
| $L_{\max}$ (report text) | 158 dB | 158 dB | 133 dB |
| $d'_{\text{BP}}$ | 452.4 m | 452.4 m | (implicit) |
| $R_{\max}$ (UMa NLOS) | 3680 m | 3680 m | 843 m |
| Reuse Factor $N$ | 1 | 7 | 7 |
| $D/R$ | 1.732 | 4.583 | 4.583 |
| Total sites (final) | 8 (5M+2$\mu$+1U) | 17 (4N+13U) | 10 (8M+1$\mu$+1U) |
| Macrocell $R$ | 800 m | 750 m | 800 m |
| Umbrella $R$ | 2500 m | 1000 m | 1000 m |
| Margin at cell edge | 25.9 dB | 27.0 dB | 0.89 dB |
| $H$ matrix dim. | $2 \times 8$ | $2 \times 4$ | $2 \times 4$ |
| CDL steps implemented | 12 | 6+discussion | 6+discussion |

## 11 Summary of Differences and Assessment

### 11.1 Strengths by Report

Table 12: Strengths and Weaknesses Summary

| Report | Strengths | Weaknesses |
|---|---|---|
| **Kagal** | • Full 12-step CDL implementation<br>• $2 \times 8$ cross-polarised MIMO<br>• Professional LaTeX formatting<br>• Street-level site photography<br>• UMi NLOS model for micro-cells<br>• Simulation output plots (PDP, APS, $\lvert H \rvert$)<br>• Independent, original code | • No GPS coordinates for sites<br>• Relatively brief iteration descriptions<br>• Does not compute thermal noise floor explicitly |
| **Ralegaon** | • Extensive iteration methodology<br>• GPS coordinates for all 17 sites<br>• Detailed umbrella cell justification<br>• KML-based mapping | • Code is identical to Pombhurna, not adapted<br>• Code says $L_{\max} = 133$ dB but report says 158 dB<br>• 13 umbrella cells is architecturally unusual<br>• No simulation output plots<br>• Steps 7–12 are discussion only |
| **Pombhurna** | • Most detailed link budget (thermal noise, NF, SNR)<br>• Explicit hand-calculations for each CDL step<br>• Frequency-dependent DS formula<br>• Clean tabular presentation of PL vs. distance | • Only 3 clusters demonstrated (vs. full 20)<br>• No simulation output plots<br>• Steps 7–12 are discussion only<br>• Very tight margin (0.89 dB) questions robustness |

### 11.2 Identical Content Alert

**Important Finding:** The Python code in the Ralegaon report (by Atharva S Chaudhari) is **identical** to the Python code in the Pombhurna report (by Prasad Patil). All variable names, comments, random seed (`np.random.seed(42)`), system parameters ($G_{\text{BS}} = 17$ dBi, $L_f = 3$ dB, $L_{\max} = 133$ dB), and even the docstring "`Pombhurna 4G Network Design`" are preserved in the Ralegaon report without modification. Furthermore, Step 12 in the Ralegaon report still references "representative of Pombhurna's semi-urban environment." The analytic sections (Part 2, Steps 1–6) in both reports are also textually identical, including the same numerical examples ($X_1 = 0.5, X_2 = 0.3, X_3 = 0.8$; same computed delays 677 ns, 1175 ns, 218 ns).

### 11.3 Overall Assessment

1. **Link Budget Approach:** Kagal and Ralegaon use a simpler extended budget ($L_{\max} = 158$ dB); Pombhurna uses a more detailed approach with explicit noise floor, body loss, and penetration loss ($L_{\max} = 133$ dB). Both approaches are valid — the choice depends on whether indoor coverage and body loss are design requirements.
2. **Network Architecture:** Kagal presents the most balanced design (5 macro + 2 micro + 1 umbrella). Ralegaon's 17-site design is over-engineered for a town of 18.5k people. Pombhurna's design is operationally tight (<1 dB margin).
3. **Channel Simulation:** Kagal is the only report with a full, end-to-end 12-step CDL implementation producing actual simulation results (PDP, APS, heatmap). The other two stop at Step 6.
4. **Code Originality:** Kagal's code is independently authored. Ralegaon and Pombhurna share identical code.